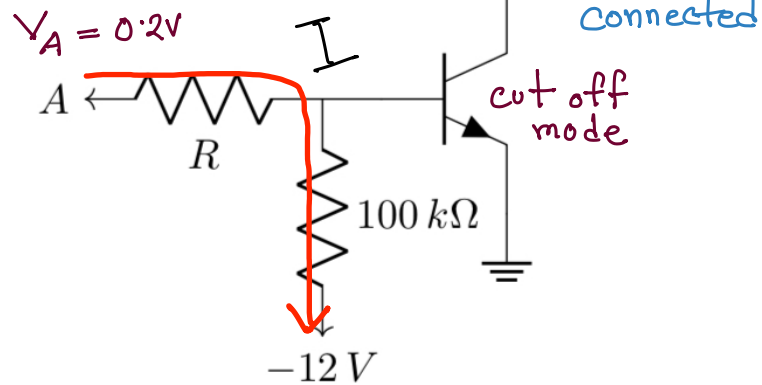


Set B

Question - 1

(a) when $A = 0 \rightarrow$

$$I = \frac{0.2 - (-12)}{100 + R} = \frac{12.2}{100 + R}$$



given, $P = \Delta V \cdot I$

$$\Rightarrow 1.24\text{ mW} = \{0.2 - (-12)\} \times \frac{12.2}{100 + R}$$

$$\Rightarrow 100 + R = \frac{12.2 \times 12.2}{1.24}$$

$$\therefore R = 120.03 - 100$$

$$\therefore R = 20.03\text{ k}\Omega$$

(b) Case: When driver's output is **Low**
(0.2V)

The OR gate loads will not take any I_L current. So, $I_L = 0$

So, we can connect as much as possible loads in this case,

$$\text{Max}^m \text{ Fanout} = \infty$$

Case: When driver's output is **High**
($V_{OH} = 10V$)

Determining $I_L \rightarrow$

$$[\because V_{OH} = 10]$$

$$\frac{10 - (V_{O2} + 0.7)}{1 \text{ k}\Omega} = \frac{V_{O2} - 0}{120 \text{ k}\Omega}$$

$$\therefore V_{O2} = 9.223 \text{ volts}$$

$$\therefore I_L = \frac{9.223 - 0}{120 \text{ k}\Omega} = 0.0768 \text{ mA}$$

So, $I_c' = N \times I_L$

$$\frac{12-10}{2 \text{ k}\Omega} = N \times (0.0768)$$

$$N = 13.02$$

$$\therefore \text{floor}(N) = 13$$

$$\text{Max}^m \text{ Fanout} = 13$$

$\therefore$ Max^m fanout considering both cases $\rightarrow$

$$\begin{aligned} \text{Max}^m \text{ Fanout} &= \min(\infty, 13) \\ &= 13 \end{aligned}$$

□ Power dissipation when only 1 Load is connected $\rightarrow$

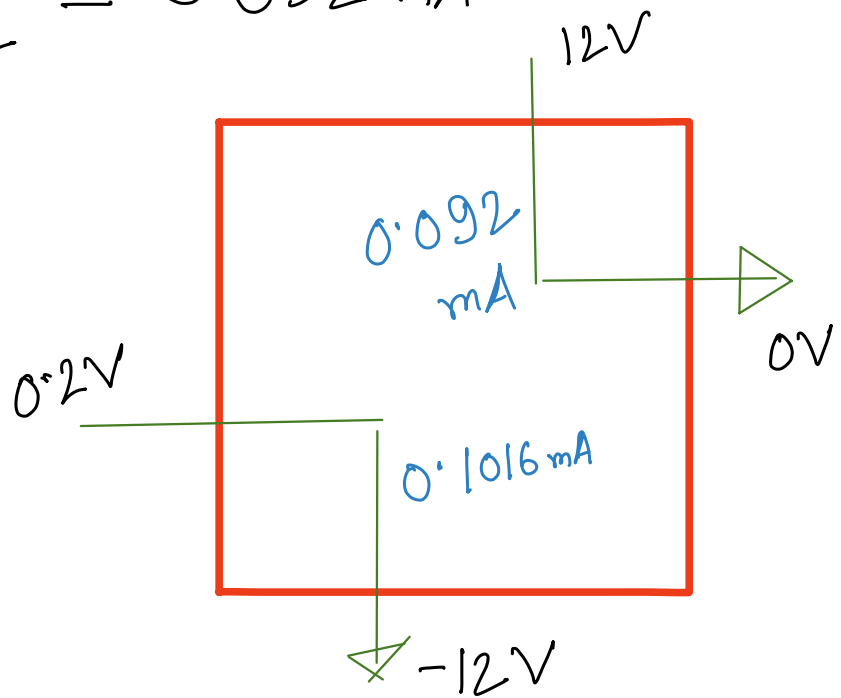
Figure out I_L $\rightarrow$
$$\frac{12 - (V_{o2} + 0.7)}{(2+1) \text{ k}\Omega} = \frac{V_{o2} - 0}{120 \text{ k}\Omega}$$

$$\therefore V_{o2} = 11.024 \text{ volts}$$

$$\therefore I_L = \frac{11.024 - 0}{120 \text{ k}\Omega} = 0.092 \text{ mA}$$

RTL cut off ($A=0$):

$$I_1 = \frac{0.2 - (-12)}{20.03 + 100} = 0.11 \text{ mA}$$



$$\therefore \text{Total Power} = [0.2 - (-12)] \times 0.1016 + [(12 - 0) \times 0.092]$$

Total Power when ' $A=0$ ' $\rightarrow$ 2.343 mW

RTL saturation ($A=1$): No load current
 $I_L = 0$

$$I_1 = \frac{12 - 0.8}{20.03 \text{ k}\Omega} = 0.56 \text{ mA}$$

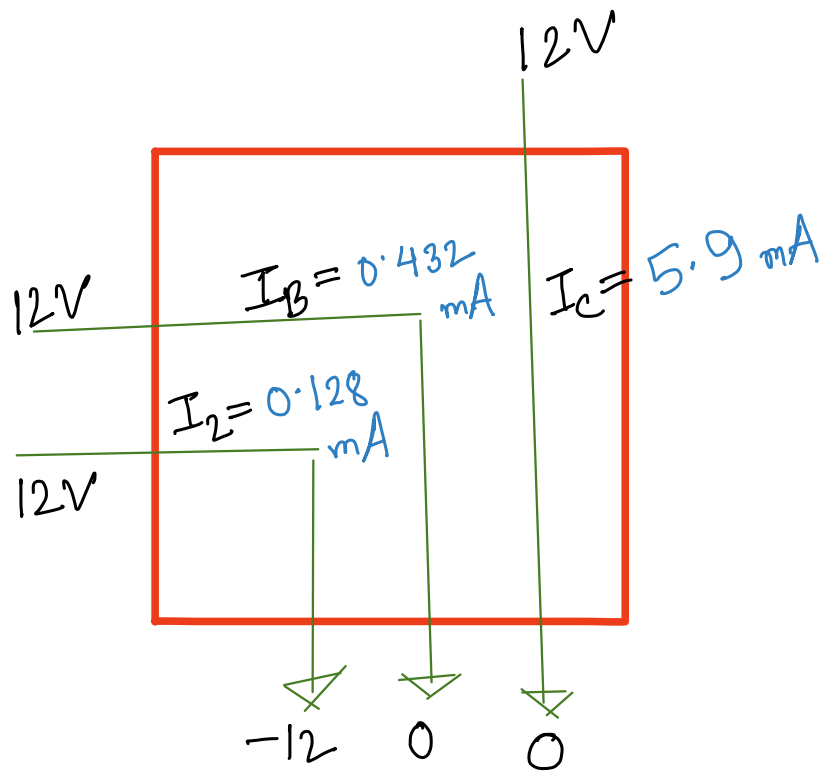
$$I_2 = \frac{0.8 - (-12)}{100 \text{ k}\Omega} = 0.128 \text{ mA}$$

$$I_B = I_1 - I_2 = 0.56 - 0.128 = 0.432 \text{ mA}$$

$$I_C = \frac{12 - 0.2}{2 \text{ k}\Omega} = 5.9 \text{ mA}$$

$\therefore$ Total Power

$$\begin{aligned} P &= P_1 + P_2 + P_3 \\ &= (12 + 12) \times 0.128 \\ &\quad + (12 - 0) \times 0.432 \\ &\quad + (12 - 0) \times 5.9 \end{aligned}$$



Total Power
when 'A = 1' $\rightarrow$ 79.056 mW

This is the Max^m power.

(c)

Given,

$$V_{OH} = 10V$$

$$V_{OL} = 0.2V$$

V_{IL} calculation:

$$\frac{V_{IL} - 0.5}{20k\Omega} = \frac{0.5 - (-12)}{100k\Omega}$$

$$\frac{V_{IL} - 0.5}{20k\Omega} = 0.125 \text{ mA}$$

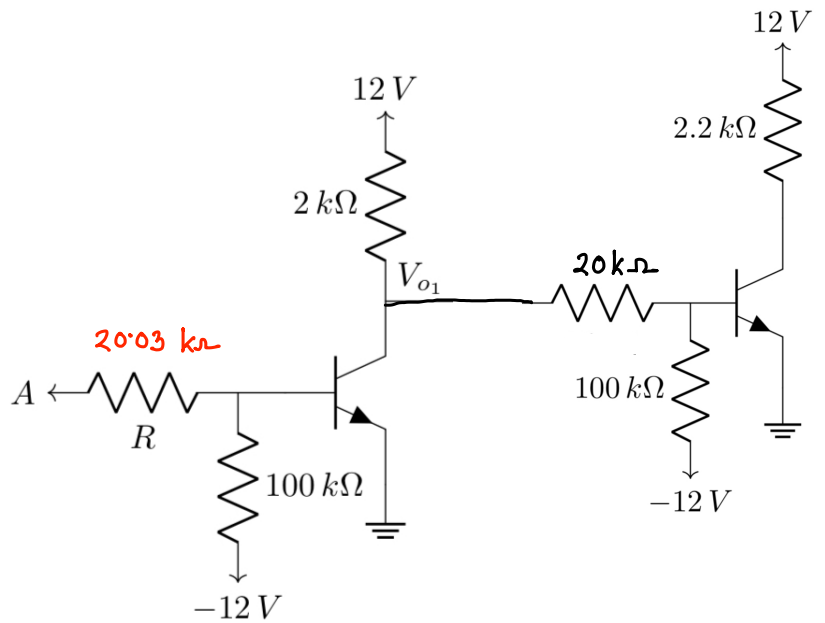
$$\therefore V_{IL} = 3 \text{ volts}$$

V_{IH} calculation:

$$I_C = \frac{12 - 0.2}{2.2k\Omega} = 5.36 \text{ mA}$$

$$\beta_F = \frac{I_C}{I_B}$$

$$\Rightarrow I_B = \frac{I_C}{\beta_F}$$



$$\therefore I_B = \frac{5.36}{25} = 0.2144 \text{ mA}$$

$$I_1 = I_2 + I_B$$

$$\Rightarrow \frac{V_{IH} - 0.8}{20 \text{ k}\Omega} = \frac{0.8 - (-12)}{100 \text{ k}\Omega} + 0.2144$$

$$\therefore V_{IH} = 7.65 \text{ volts}$$

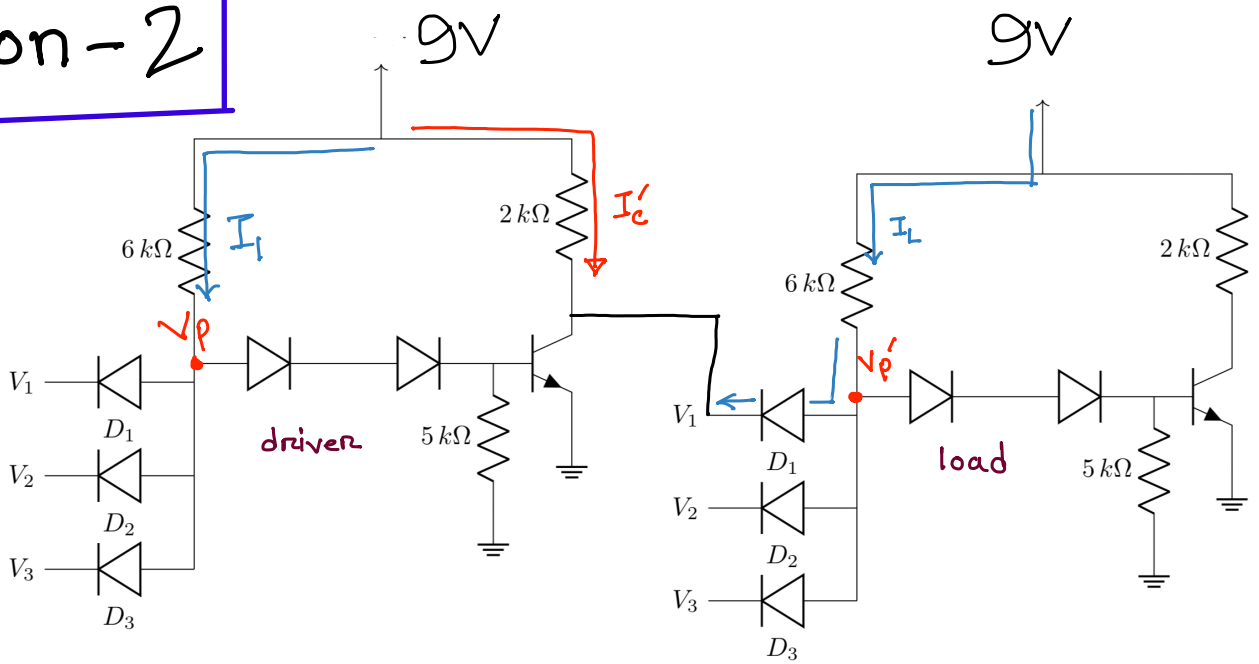
$$\therefore \boxed{V_{NH}} = V_{OH} - V_{IH} = 10 - 7.65 \\ = 2.35 \text{ volts}$$

$$\therefore \boxed{V_{NL}} = V_{IL} - V_{OL} = 3 - 0.2 \\ = 2.8 \text{ volts}$$

$$\therefore \text{Noise margin} = \min(V_{NH}, V_{NL}) \\ NM = \min(2.35, 2.8) \\ NM = 2.35 \text{ volts}$$

Question-2

(a)



Case: When driver output is Low (0.2V)

$$V_{p'} = 0.2 + 0.7 = 0.9 \text{ volts}$$

$$\therefore \text{Individual load current, } I_L = \frac{9 - 0.9}{6k\Omega} = 1.35 \text{ mA}$$

$$I_{c'} = \frac{9 - 0.2}{2k\Omega} = 4.4 \text{ mA}$$

$$\text{Now, } I_{c_{\max}} = \beta_F \times I_B$$

$$= 25 \times [I_1 - I_2]$$

$$= 25 \times \left[\frac{9 - 2.2}{6k\Omega} - \frac{0.8 - 0}{5k\Omega} \right]$$

$$\therefore I_{c_{max}} = 24.33 \text{ mA}$$

$$\text{Now, } I_{c_{max}} = I_c' + N \times I_L$$

$$\Rightarrow 24.33 = 4.4 + N \times 1.35$$

$$\Rightarrow N = \frac{24.33 - 4.4}{1.35}$$

$$\therefore N = 14.763$$

$\therefore \text{floor}(N) = 14 \rightarrow \text{Max}^m \text{ Fanout for this case}$

Case: When driver's output is **High**
($V_{OH} = 9V$)

due to high output, every diode will be in Reverse biased, So, diode current on the current flowing to driven side will be zero,
 $\therefore I_L = 0 \text{ mA}$

Thus we can connect as many loads as possible,

Max^m fanout for this case $\rightarrow N = \infty$

$\therefore$ Max^m Fanout considering both cases $= \min(14, \infty)$

$$= 14$$

(b)

Driven input 'high' $\rightarrow$ any of the inputs 'low'

if all input Low $\rightarrow V_1 = V_2 = V_3 = 0V$
then all diodes D_1, D_2, D_3 will be Forward biased,

$$\therefore I_{D_1} = I_{D_2} = I_{D_3} = \frac{I_1}{3}$$

$$= \frac{\left(\frac{9 - 0.7}{6k\Omega} \right)}{3}$$

[input voltage is 0V for Low]



$$[V_p = 0 + 0.7 = 0.7 \text{ volt}]$$

$$= 0.4611 \text{ mA}$$

(c)

$$I_{C3} = \frac{6 - 0.2}{4k\Omega} = 1.45 \text{ mA}$$

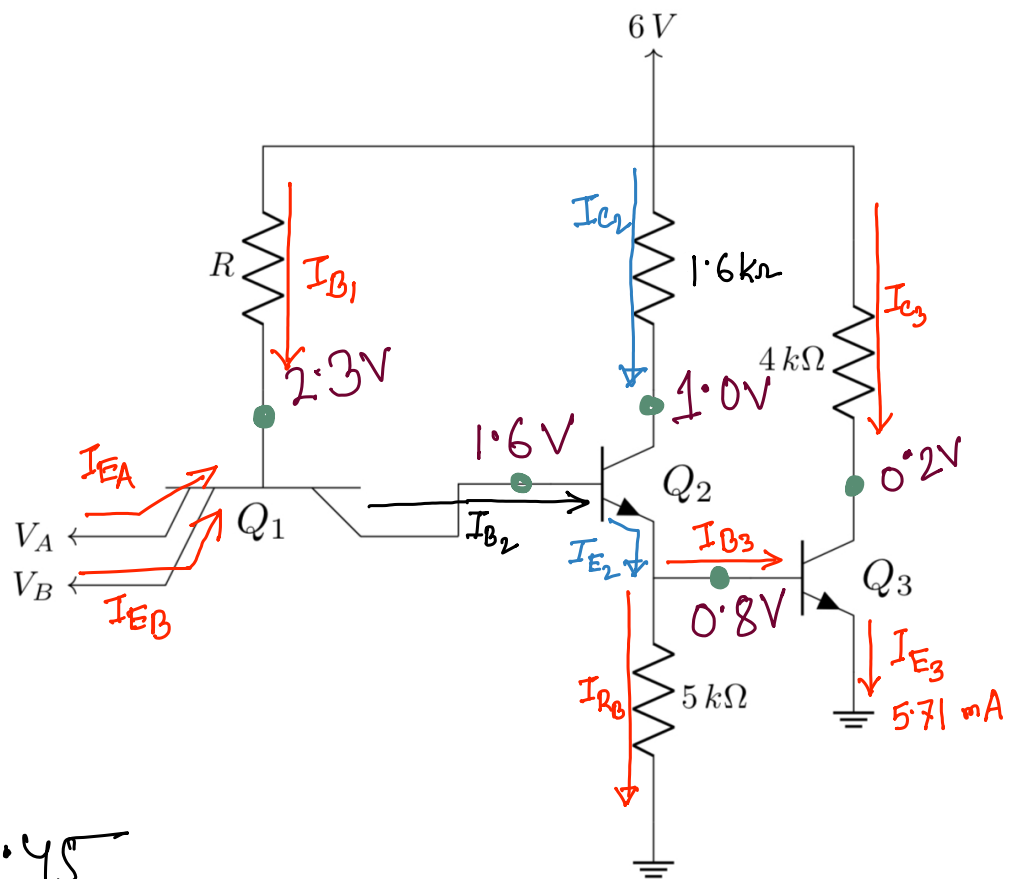
$$I_{B3} = I_{E3} - I_{C3} = 5.71 - 1.45 = 4.26 \text{ mA}$$

$$I_{E2} = I_{B3} + I_{R_B} = 4.26 + \left(\frac{0.8 - 0}{5k\Omega} \right) = 4.42 \text{ mA}$$

$$I_{C2} = \frac{6 - 1}{1.6k\Omega} = 3.125 \text{ mA}$$

$$I_{B2} = I_{E2} - I_{C2} = 4.42 - 3.125 = 1.295 \text{ mA}$$

$$I_{E_A} = I_{E_B} = (\beta_R \times I_{B1}) = 0.2 \times \left(\frac{6 - 2.3}{R} \right)$$



$$I_{B_2} = I_{B_1} + \underbrace{I_{E_A} + I_{E_B}}$$

$$1.295 = \left(\frac{6 - 2.3}{R} \right) + 2 \times 0.2 \times \left(\frac{6 - 2.3}{R} \right)$$

$$\Rightarrow 1.295 = 1.4 \times \left(\frac{6 - 2.3}{R} \right)$$

$$\therefore R = 1.4 \times \left(\frac{6 - 2.3}{1.295} \right)$$

$$\therefore R = 4 \text{ k}\Omega$$

Question-3

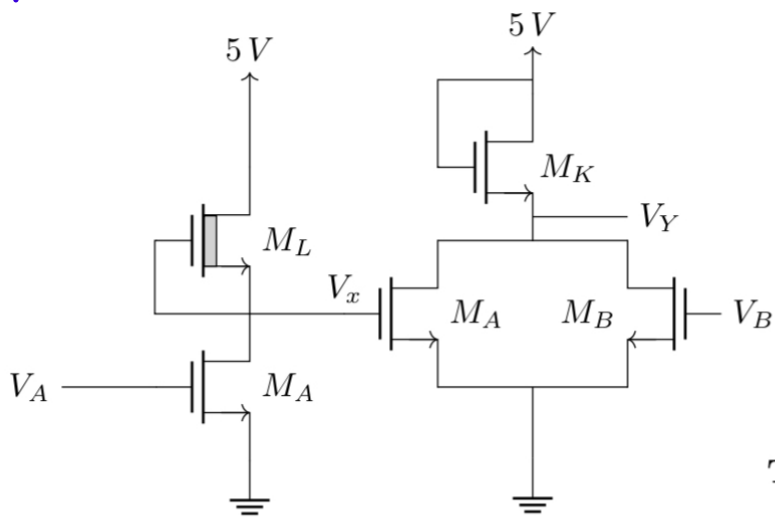


Figure 5

	$K_n \left(\frac{\text{mA}}{\text{V}^2} \right)$	$V_{TN} \text{ (V)}$
M_L	0.5	-0.4
M_A	0.65	0.5
M_B	0.75	0.6
M_K	0.3	1

Table 1: MOSFET parameters

Triode Mode: $I_{DS} = K_n[2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2]$

Saturation Mode: $I_{DS} = K_n(V_{GS} - V_{TN})^2$

(a) When $V_A = 5$ volts (high)
 M_A will be in triode mode (assumption)
 and, M_L will be in saturation mode,

$$I_{M_A \text{ triode}} = I_{M_L \text{ sat}}$$

$$V_{GSL} = V_{GL} - V_{SL} = 0$$

$$\Rightarrow 0.65 \times [2(5 - 0.5)V_n - V_n^2] = 0.5 \times [0 - (-0.4)^2]$$

$$\Rightarrow 9V_n - V_x^2 - 0.123 = 0$$

$$\therefore V_x = 0.0137 \text{ volts}$$

Now, $V_{DS} < V_{GS} - V_{TN}$

$$\Rightarrow 0.0137 < (5 - 0.5)$$

so, M_A is triode mode $\rightarrow$ verified

Now, $V_X = 0.0137$ volts (low) $\rightarrow M_A$ cut off (NOT)

$V_B = 5$ volts (high) $\rightarrow M_B$ triode (assume)

and, M_K in saturation.

So,

$$I_{M_B \text{ triode}} = I_{M_K \text{ sat}}$$

$$V_{GSL} = V_{GL} - V_{SL}$$

$$\Rightarrow 0.75 [2(5 - 0.6)V_y - V_y^2] = 0.3 [(5 - V_y) - 1]^2$$

$$\Rightarrow 8.8V_y - V_y^2 = \frac{0.3}{0.75} [4 - V_y]^2$$

$$\begin{aligned} \Rightarrow 8V_y - V_y^2 &= 0.4 \times [16 - 8V_y + V_y^2] \\ &= 6.4 - 3.2V_y + 0.4V_y^2 \end{aligned}$$

$$\Rightarrow 1.4V_y^2 - 12.2V_y + 6.4 = 0$$

$$\therefore \boxed{V_y = 0.56} \text{ and } 8.153 \text{ volts}$$

↓
unacceptable

Now, $V_{DS} < V_{GS} - V_{TN}$

$$\Rightarrow 0.56 < (5 - 0.6)$$

so, M_B is triode mode $\rightarrow$ verified

(b)

current flowing through ' M_L ' $\rightarrow$

$$I_{M_L \text{ sat}} = 0.5 [0 - (-0.4)]^2$$

$$= 0.08 \text{ mA}$$

$$\therefore \underline{P_{M_L \text{ sat}}} = (5 - 0.0137) \times 0.08$$

$$= 0.3989 \text{ mW}$$

$$\approx \boxed{0.4 \text{ mW}}$$

current flowing through $M_k \rightarrow$

$$I_{M_k \text{ sat}} = 0.3 \left[(5 - 0.56) - 1 \right]^2$$
$$= 3.55 \text{ mA}$$

$$\therefore \underline{P_{M_k \text{ sat}}} = (5 - 0.56) \times 3.55$$
$$= \boxed{15.762 \text{ mW}}$$

(c)

Boolean logic function

CMOS implementation

$$Y = \overline{A+B}$$

$$= \overline{A} \cdot \overline{B}$$

$$= A \cdot \overline{B}$$

